

CLASS 10 · ICSE · SELINA CONCISE MATHEMATICS · SHORT NOTES

# Chapter 13: Section and Mid-Point Formula

Topic-wise short notes with solved examples

These formulas give the coordinates of a point that divides a line segment in a given ratio. The mid-point and the centroid of a triangle are special cases.

## 1 The section formula

- Let  $P(x, y)$  divide the line segment joining  $A(x_1, y_1)$  and  $B(x_2, y_2)$  in the ratio  $m_1 : m_2$ , so that  $AP : PB = m_1 : m_2$ . Then

$$x = \frac{m_1x_2 + m_2x_1}{m_1 + m_2} \quad y = \frac{m_1y_2 + m_2y_1}{m_1 + m_2}$$

- Each part of the ratio multiplies the coordinates of the **opposite** end:  $m_1$  goes with  $B$  and  $m_2$  goes with  $A$ .

**Example 1.1** Find the coordinates of the point that divides the segment joining  $A(2, -1)$  and  $B(7, 9)$  in the ratio  $3 : 2$ .

SOLUTION

$$x = \frac{3 \times 7 + 2 \times 2}{3 + 2} = \frac{25}{5} = 5$$

$$y = \frac{3 \times 9 + 2 \times (-1)}{3 + 2} = \frac{25}{5} = 5$$

The point is  $(5, 5)$ .

## 2 Finding the ratio

- To find the ratio in which a point divides a segment, take the ratio as  $k : 1$  and use one coordinate to find  $k$ .
- A point on the  $x$ -axis has ordinate 0. A point on the  $y$ -axis has abscissa 0.
- If  $k$  comes out negative, the point divides the segment externally.

**Example 2.1** In what ratio does the point  $P(3, 4)$  divide the segment joining  $A(1, 2)$  and  $B(6, 7)$ ?

SOLUTION

Let the ratio be  $k : 1$ . Then  $\frac{6k + 1}{k + 1} = 3$ , so  $6k + 1 = 3k + 3$  and  $k = \frac{2}{3}$ .

Check with  $y$ :  $\frac{7k + 2}{k + 1} = \frac{\frac{14}{3} + 2}{\frac{5}{3}} = 4$ . Correct.

The ratio is  $2 : 3$ .

**Example 2.2** In what ratio does the  $y$ -axis divide the segment joining  $A(-4, 5)$  and  $B(6, 0)$ ? Find the point of intersection.

**SOLUTION**

Let the ratio be  $k : 1$ . On the  $y$ -axis the abscissa is 0:  $\frac{6k - 4}{k + 1} = 0$ , so  $k = \frac{2}{3}$ .

The ratio is 2 : 3. Ordinate =  $\frac{2 \times 0 + 3 \times 5}{2 + 3} = 3$ .

**Ratio 2 : 3; the point is (0, 3).**

### 3 Points of trisection

- The points of trisection divide a segment into three equal parts. They divide it in the ratios 1 : 2 and 2 : 1.

**Example 3.1** Find the points of trisection of the segment joining  $A(2, 1)$  and  $B(8, 10)$ .

**SOLUTION**

Ratio 1 : 2:  $P = \left( \frac{8 + 4}{3}, \frac{10 + 2}{3} \right) = (4, 4)$

Ratio 2 : 1:  $Q = \left( \frac{16 + 2}{3}, \frac{20 + 1}{3} \right) = (6, 7)$

**The points are (4, 4) and (6, 7).**

### 4 The mid-point formula

- The mid-point of the segment joining  $(x_1, y_1)$  and  $(x_2, y_2)$  is

$$\left( \frac{x_1 + x_2}{2}, \frac{y_1 + y_2}{2} \right)$$

- If one end and the mid-point are known, the other end can be found.
- The diagonals of a parallelogram bisect each other, so they have the same mid-point.

**Example 4.1** The mid-point of the segment joining  $A(3, -2)$  and  $B$  is  $M(1, 4)$ . Find  $B$ .

**SOLUTION**

Let  $B = (x, y)$ . Then  $\frac{3 + x}{2} = 1$  and  $\frac{-2 + y}{2} = 4$ .

**$B = (-1, 10)$**

**Example 4.2**  $A(2, 3)$ ,  $B(6, 5)$  and  $C(8, 9)$  are three vertices of a parallelogram  $ABCD$ . Find  $D$ .

**SOLUTION**

Mid-point of  $AC = (5, 6)$ . This is also the mid-point of  $BD$ .

Let  $D = (x, y)$ :  $\frac{6+x}{2} = 5$  and  $\frac{5+y}{2} = 6$

$D = (4, 7)$

**Example 4.3**  $AB$  is a diameter of a circle with centre  $C(2, -3)$ . If  $A = (5, 1)$ , find  $B$ .

**SOLUTION**

The centre is the mid-point of the diameter:  $\frac{5+x}{2} = 2$  and  $\frac{1+y}{2} = -3$ .

$B = (-1, -7)$

## 5 Centroid of a triangle

- A **median** joins a vertex to the mid-point of the opposite side. The three medians meet at the **centroid**  $G$ .
- The centroid divides each median in the ratio  $2 : 1$ , measured from the vertex.

$$G = \left( \frac{x_1 + x_2 + x_3}{3}, \frac{y_1 + y_2 + y_3}{3} \right)$$

**Example 5.1** Find the centroid of the triangle with vertices  $A(1, 4)$ ,  $B(-2, -1)$  and  $C(7, 3)$ .

**SOLUTION**

$$G = \left( \frac{1 - 2 + 7}{3}, \frac{4 - 1 + 3}{3} \right)$$

$G = (2, 2)$

**Example 5.2** The centroid of a triangle is  $(1, 3)$  and two of its vertices are  $(2, 5)$  and  $(-4, 1)$ . Find the third vertex.

**SOLUTION**

$\frac{2 - 4 + x}{3} = 1$  gives  $x = 5$ , and  $\frac{5 + 1 + y}{3} = 3$  gives  $y = 3$ .

The third vertex is  $(5, 3)$ .

## 6 Common mistakes

- Multiplying  $m_1$  by  $x_1$ . It multiplies  $x_2$ .
- Writing the ratio the wrong way round:  $AP : PB$  is not the same as  $PB : AP$ .
- Forgetting that a point on the  $x$ -axis has  $y = 0$ , not  $x = 0$ .
- Using the mid-point formula with a subtraction instead of an addition.

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